

BUILT FOR
NABARD
HACKATHON
@ GLOBAL
FINTECH
FEST 2026

GramOS

AI-Powered
Rural

Financial
Intelligence
Platform
Predicting
financial
stress

before it
becomes
financial
failure.

The Foundation

Problem & Why Now

The financial health of India's rural micro-enterprises is monitored manually, leading to delayed interventions and rising NPAs.

Why Now: The EASE 9.0 reforms launched by the Government of India in February 2026 mandate technology-led modernization, AI integration, and operational excellence for public sector banks. Combined with the rapid expansion of rural digital payments (UPI), the data ecosystem required to proactively prevent defaults is now fully accessible.

The Solution

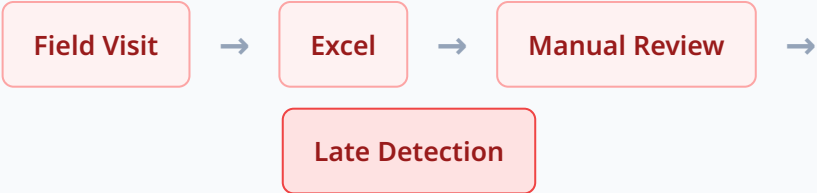
GramOS is an AI-powered Rural Financial Intelligence Platform that enables financial institutions to proactively monitor the financial health of rural micro-enterprises using alternative data and explainable AI.

The platform combines digital payment signals (UPI), commodity prices, weather, seasonal patterns, and historical financial data to forecast cash flows over a 3–6 month horizon, identify early signs of financial stress, and recommend timely interventions. Unlike traditional black-box models, GramOS provides transparent, explainable insights that help field officers and financial institutions make informed lending and support decisions.

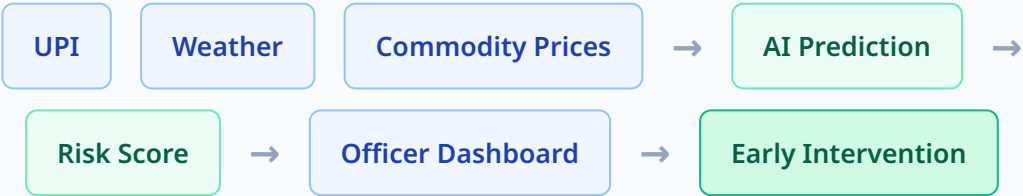
System Evolution

Current System vs. GramOS Architecture

Current Reactive Model:



GramOS Proactive Intelligence:



Process Flow & Architecture

1. Data Collection

Collect alternative data including UPI transaction trends, commodity prices, climate information, seasonal indicators, and available financial records.



2. Data Intelligence Layer

Integrate structured and unstructured data into a unified intelligence layer to build contextual profiles for each rural micro-enterprise. *(Powered by PostgreSQL with pgvector).*



3. AI Prediction Engine

Forecast cash flows over a 3–6 month horizon and identify potential financial stress using predictive AI models.



4. Explainable Risk Analysis

Generate transparent explanations highlighting the factors contributing to the predicted risk.



5. Recommendation Engine & Decision Dashboard

Recommend actionable interventions (e.g., monitoring, restructuring) and present forecasts through an intuitive dashboard for field officers.

Explainable AI (XAI) in Action

Enterprise: Village Dairy

Risk: HIGH

Confidence 94%

Cash Flow Forecast:

↓ 21%

Why?

- ✓ Feed prices +18%
- ✓ Milk price ↓
- ✓ UPI inflow ↓ 12%
- ✓ Seasonal demand shifts

Recommendation: Offer Working Capital • Review in 7 days

Commercial Potential & Impact

Commercial Potential

GramOS will operate as a B2B SaaS platform and AI infrastructure layer for NABARD, Regional Rural Banks (RRBs), Cooperative Banks, NBFCs, and Agri-FinTechs. The platform can be licensed on an enterprise subscription basis with optional usage-based pricing for analytics and AI services.

Expected Impact

- Designed to forecast cash flows over a 3–6 month horizon using multi-source alternative data.
- Earlier identification of financially stressed enterprises.
- Improved credit monitoring using alternative data.
- Reduced manual effort for field officers through intelligent prioritization.
- Better explainability and transparency in credit risk assessment.
- Stronger support for financial inclusion through proactive interventions.

**Model performance and exact outcome metrics will be validated through initial pilot deployments.*

Team

Sudarshan Kulkarni

Founder & AI Systems Engineer

- Built 5 production AI systems
- Production Voice AI
- Production RAG
- FinTech AI
- FastAPI Integration
- LangGraph Orchestration
- Real-time AI Infrastructure